INFOSYS VIRTUAL INTERNSHIP 6.0

PREFERENCES FOOD TRENDS UNDERSTANDING CUSTOMER IN FB

BATCH: 11 | GROUP : 2 | TEAM:C

YEAR: 2026

Internship Duration: 8 Weeks

Mentor:

Jayasri Vangala Mam

Submitted By:

Rekha Shri Giduthuri

Rohit Ranjan

Naresh Kumar

Nasina Asritha

Nithya Thevath

Naitik Kanojiya

Gummalla Nagalakshmi

Infosys Springboard Virtual Internship 6.0 Completion Report

1.Project Title:-

Food Trends Understanding Customer Trend in Facebook

2. Project execution details;-

This project focuses on analyzing customer trends in food brands by integrating structured data such as customer demographics, product purchases, ratings, and purchase frequency with unstructured data like social media comments, app reviews, and customer feedback. The main goal of the project is to understand customer behavior, identify satisfaction levels, and discover factors that influence customer loyalty and revenue growth. Using analytical techniques such as sentiment analysis, product analysis, and customer segmentation, the project converts raw data into meaningful insights. All the results are visualized through an interactive Power BI dashboard to support data-driven business decisions.

Key Points:

1. Collected structured data such as customer demographics, product purchases, ratings, and purchase frequency.

2. Integrated unstructured data including social media comments, reviews, and open-text feedback. Cleaned and organized the data using a data modeling framework (Star Schema).

3. Applied sentiment analysis to classify feedback into positive, neutral, and negative categories. Performed product insight analysis to understand revenue contribution and product performance.

4. Used customer segmentation to identify loyal, neutral, and at-risk customers.

5. Built an interactive Power BI dashboard to visualize trends and key performance indicators. Enabled data-driven decision making to improve customer satisfaction and business growth

Project Approach:-

The project followed a structured and systematic approach to analyze restaurant data and understand customer preferences. Initially, the dataset was collected from Kaggle and examined to understand its structure and attributes. Data cleaning and preprocessing were performed using Power Query to remove missing values, duplicates, and inconsistencies. After preparing the dataset, a data model was created in Power BI to establish relationships and support analysis. Interactive dashboards were then developed using various visualizations and filters to present insights clearly. Finally, the dashboards were analyzed to identify trends, patterns, and meaningful insights related to customer behavior and restaurant performance. Key Points Followed a structured data analytics approach Collected and explored dataset Cleaned and preprocessed data Built a data model in Power BI Created interactive dashboards Generated insights from the dashboards

Project Steps:-

Step 1: Data Collection Paragraph

The first step involved collecting the dataset from Kaggle, which contained detailed information about restaurants such as location, cuisine type, pricing, ratings, vote counts, and service availability. The dataset was imported into Power BI for further analysis. Key Activities Selected dataset from Kaggle Imported data into Power BI Reviewed dataset structure and attributes.

Step 2: Data Cleaning and Preprocessing

The dataset contained missing values and inconsistencies that needed to be corrected before analysis. Using Power Query, missing data was handled, duplicate records were removed, and categorical fields were standardized to ensure consistency and accuracy. Key Activities Handled missing values Removed duplicate records Standardized and formatted data.

Step 3: Data Modeling

After cleaning the data, the dataset was structured properly to support efficient analysis. Data relationships were established, and calculated measures such as average ratings, total votes, and restaurant counts were created using DAX. Key Activities Organized data structure Created calculated measures and KPIs Built relationships between tables.

Step 4: Dashboard Development

Interactive dashboards were developed in Power BI to visualize customer preferences and restaurant trends. Different types of charts, graphs, and slicers were used to make the dashboards interactive and easy to understand. Key Activities Designed multiple dashboards Added charts and visualizations Implemented filters and slicers.

Step 5: Insight Generation

The final step involved analyzing the dashboards to identify trends and patterns related to restaurant performance and customer behavior. The insights were validated through team discussions to ensure accuracy. Key Activities Analyzed trends and patterns Generated meaningful insights Validated findings with the team.

Technologies Used:-

1. Power BI: Served as the primary platform for creating interactive dashboards and visualizations, transforming raw data into actionable insights.

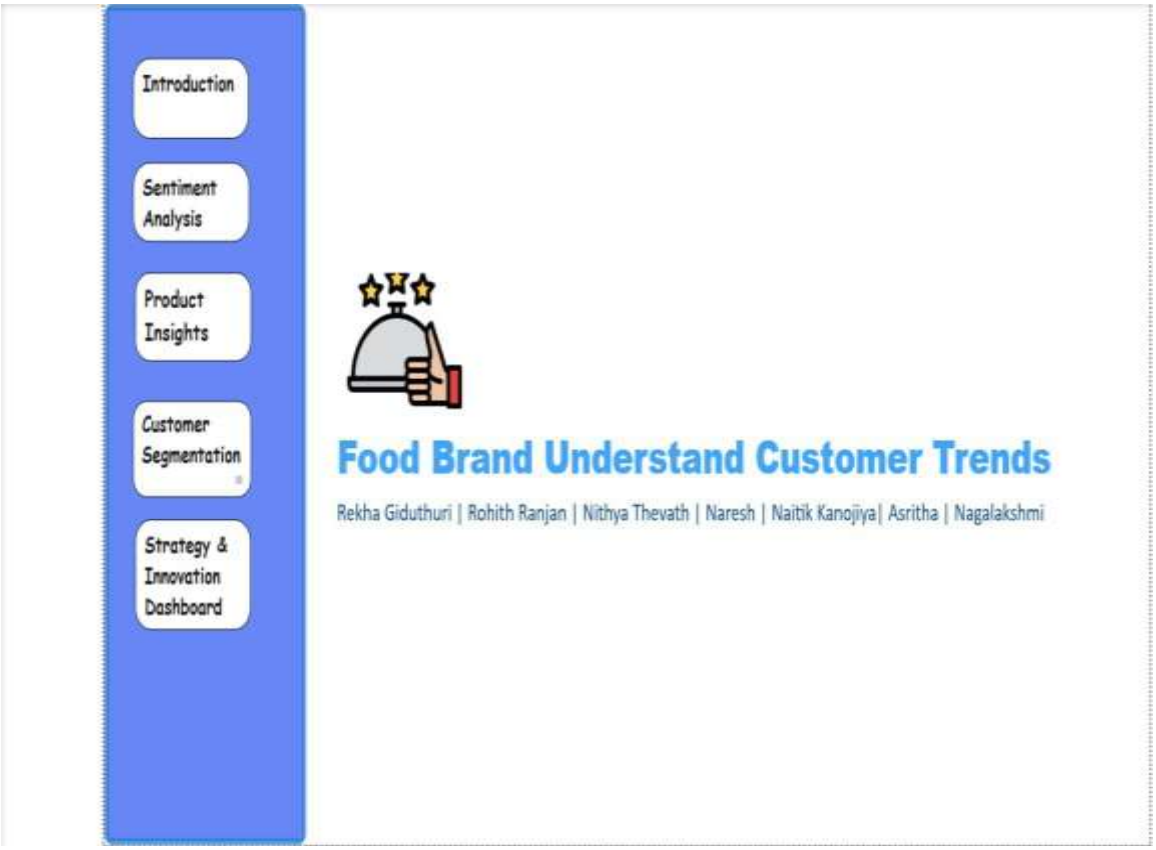
2. Power Query: Used extensively within Power BI and Excel for efficient data cleaning, shaping, and transformation processes.

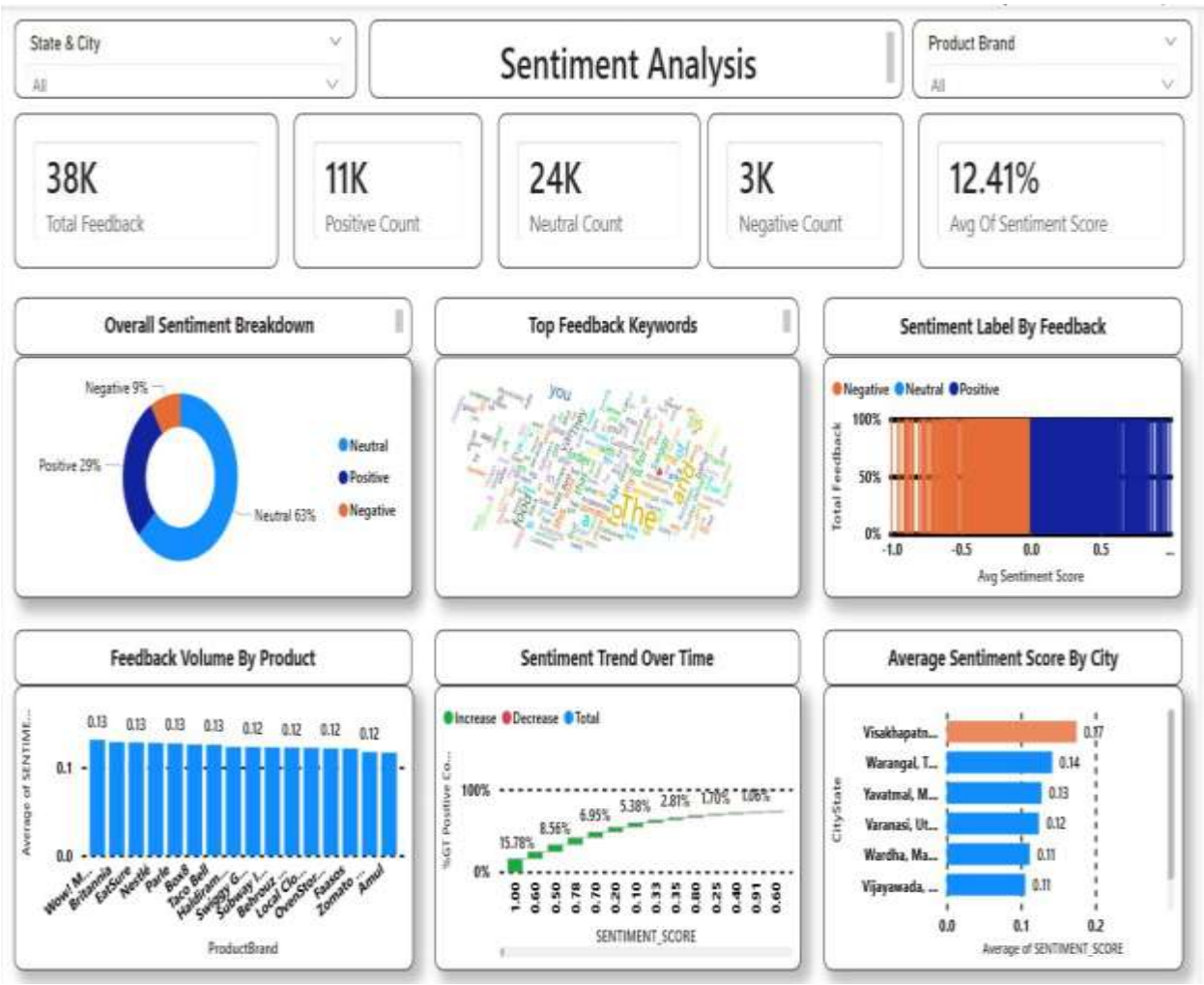
3. DAX: Employed for advanced calculations, including creating custom measures and key performance indicators (KPIs) within Power BI models.

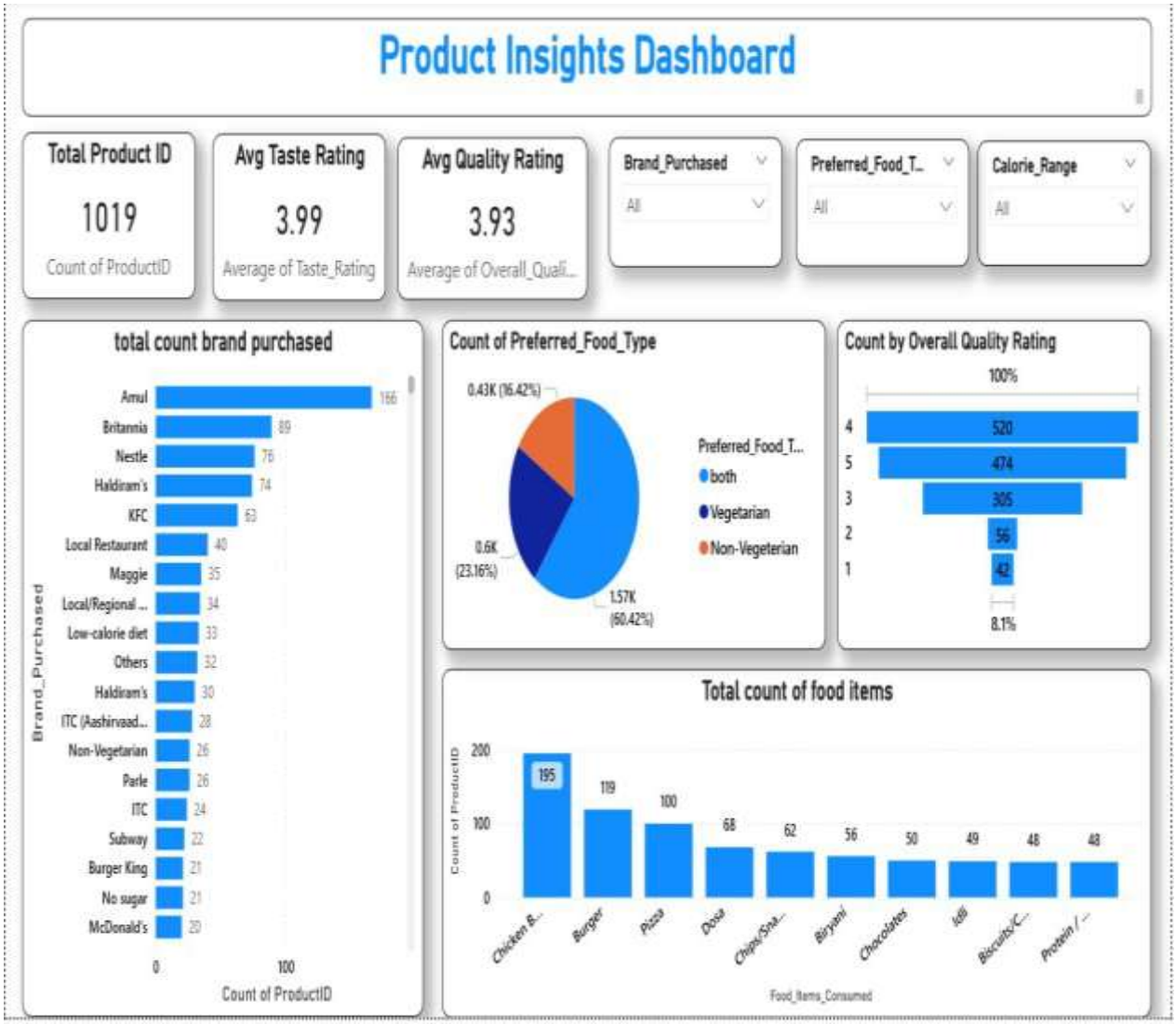
4. Microsoft Excel: Provided initial support for quick data inspection, validation, and foundational analysis before platform migration.

5. Documentation Tools: Microsoft PowerPoint and Word were used for presentation and formal project report documentation, respectively.

3. Snapshots / Screenshots:-




State & City
All
Sentiment Analysis
Product Brand
All
38K
Total Feedback
11K
Positive Count
24K
Neutral Count
3K
Negative Count
12.41%
Avg Of Sentiment Score
Overall Sentiment Breakdown
Negative 9%
Positive 29%
Neutral 63%
Neutral
Positive
Negative
Top Feedback Keywords
Sentiment Label By Feedback
Negative Neutral Positive
100%
50%
0%
Total Feedback
-1.0
-0.5
0.0
0.5
Avg Sentiment Score
Feedback Volume By Product
Average of SENTIME...
0.13 0.13 0.13 0.13 0.12 0.12 0.12 0.12
0.1
0.0
Wow! M...
Britannia
EatSure
Nestlé
Parle
Box8
Taco Bell
Haldiram...
Swiggy G...
Subway I...
Behrouz ...
Local Clo...
OvenStor...
Faasos
Zomato ...
Amul
ProductBrand
Sentiment Trend Over Time
Increase Decrease Total
%GT Positive Co...
100%
0%
15.78%
8.56%
6.95%
5.38%
2.81%
1.70%
1.06%
1.00 0.60 0.50 0.78 0.70 0.20 0.10 0.33 0.35 0.80 0.25 0.40 0.91 0.60
SENTIMENT_SCORE
Average Sentiment Score By City
CityState
Visakhapatn... 0.17
Warangal, T... 0.14
Yavatmal, M... 0.13
Varanasi, Ut... 0.12
Wardha, Ma... 0.11
Vijayawada, ... 0.11
0.0
0.1
0.2
Average of SENTIMENT_SCORE


Product Insights Dashboard
Total Product ID
1019
Count of ProductID
Avg Taste Rating
3.99
Average of Taste_Rating
Avg Quality Rating
3.93
Average of Overall_Quali...
Brand_Purchased
All
Preferred_Food_T...
All
Calorie_Range
All
total count brand purchased
Amul 166
Britannia 89
Nestle 76
Haldiram's 74
KFC 63
Local Restaurant 40
Maggie 35
Local/Regional ... 34
Low-calorie diet 33
Others 32
Haldiram's 30
ITC (Aashirvaad... 28
Non-Vegetarian 26
Parle 26
ITC 24
Subway 22
Burger King 21
No sugar 21
McDonald's 20
Brand_Purchased
0
100
Count of ProductID
Count of Preferred_Food_Type
0.43K (16.42%)
0.6K (23.16%)
1.57K (60.42%)
Preferred_Food_T...
both
Vegetarian
Non-Vegeterian
Count by Overall Quality Rating
100%
4 520
5 474
3 305
2 56
1 42
8.1%
Total count of food items
Count of ProductID
200
100
0
195
119
100
68
62
56
50
49
48
48
Chicken &...
Burger
Pizza
Dosa
Chips/Sna...
Biryani
Chocolates
Idli
Biscuits/C...
Protein / ...
Food_Items_Consumed

Customer Segmentation & Personas
Purchase_Channel
All
City
All
Brand_P...
All
Age Group
25-34
45-54
Under 25
35-44
55+
Total Customers
3.948K
total customers
Avg Rating
3.93
Avg Rating
Avg Sentiment Score
0.33
Average of...
Positive %
55.08%
Positive %
Loyal fan %
0.63
Loyal Fan %
Customer Distribution by Persona
0.54K (13.73%)
0.62K (15...)
2.49K (62.94%)
Customer Per...
Loyal Fan
Silent User
Positive User
At Risk Cus...
Customer Distribution by Age Group and Persona
Customer Persona At Risk Customer Loyal Fan Positive User Silent User
2K
1K
0K
Count of CustomerID
Under 25
25-34
35-44
45-54
55+
Age Group
Average Rating by Customer Persona
100%
Loyal Fan
4.43
Positive User
3.00
At Risk Custo...
1.51
34.1%
Customer Sentiment vs Rating Analysis
Customer Persona At Risk Customer Loyal Fan Positive User Silent User
Avg Rating
4
2
0.15
0.20
0.25
0.30
0.35
0.40
0.45
Avg Sentiment
Purchase Channel Preference by Customer Persona
Customer Persona At Risk Customer Loyal Fan Positive User Silent User
400
200
0
Count of CustomerID
Online Store
Dining
Store
Others
Dining (Restau...
Delivery Partner
Delivery Partner...
Store Purchase
Delivery
Partner Store
Purchase_Channel

38K
Total Feedback
Strategy And Innovation Dashboard
Sentiment Category Filter
Negative
Neutral
Positive
Top 10 Keywords by Feedback Volume
Yummy 1.2K
Very nice 0.6K
Yummy y... 0.4K
Nice 0.3K
Good mo... 0.2K
0.2K
Wow 0.2K
Delicious 0.2K
Keywords
0K
1K
Total Feedback
Customer Sentiment by Key Product Features
100%
50%
0%
Positive %
100.00%
Packaging & Presentation
Food Quality & Freshness
Taste & Flavor
Price & Value
Other
Keyword Category
Total Customer Feedback by Product
Zomato Kit... 118
Wow! Momo 27
Vegetarian, ... 19
Vegetarian, ... 17
Vegetarian, ... 17
Vegetarian, ... 11
Yippee 8
Vijaya Dairy 6
Brand_Purchased
0
50
100
Total Feedback
Product Sentiment by Taste Rating
Taste_Rating 1 2 3 4 5
Average of Sentim...
4
2
0
-2
0.9
0.8
1.0
1.0
Vijaya Dairy
Yippee
Zomato Kitc...
Vegetarian, ...
Vegetarian, ...
Wow! Momo
Vegan, Vege...
Vegetarian, ...
Vegetarian, ...
Vegetarian
Brand_Purchased
Monthly Customer Activity by Gender
Female
Male
Other
Prefer not to say
Count o...
100
0
83
1
11
Apr 2026
Jul 2026
Oct 2026
Timestamp

Conclusion
Overall sentiment remains **positive (55%)**, with an **average score of (12%)**, indicating stable brand perception.
Loyal customers represent the largest segment and contribute significantly higher ratings, reinforcing strong customer retention.
A small but important segment of at-**risk customers** requires targeted engagement strategies.
Product-level sentiment differences highlight clear opportunities for **quality** and **experience** improvement.
Online channels and younger demographics drive higher engagement, suggesting **strong** digital **growth** potential.

4. Challenges Faced:-

During the development of the project, several technical and analytical challenges were encountered. These included handling missing and inconsistent data entries, managing relatively large datasets within Power BI, selecting meaningful performance indicators, and interpreting rating trends where vote counts were low. Each of these challenges required careful analysis and structured problem-solving to ensure the reliability and clarity of the final dashboards.

Key Challenges and Resolutions:

Challenge 1: Inconsistent and Missing Data Resolution: Applied systematic data cleaning and preprocessing techniques using Power Query. Missing values were handled appropriately, inconsistent entries were standardized, and unnecessary columns were removed to improve data quality and consistency.

Challenge 2: Identifying Meaningful KPIs Resolution: Carefully analyzed business requirements to determine relevant metrics such as total revenue, order frequency, average ratings, and consistency scores. DAX measures were created to accurately calculate and represent these KPIs.

Challenge 3: Dashboard Clutter and Readability Issues Resolution: Improved dashboard layout by reducing redundant visuals, aligning elements properly, and incorporating slicers for dynamic filtering. This enhanced user experience and ensured better interpretability of insights.

Challenge 4: Performance Optimization in Power BI Resolution: Optimized DAX calculations, minimized redundant columns, and improved data model relationships to enhance dashboard performance and reduce loading time.

5. Learnings & Skills Acquired:-

This project enhanced practical knowledge of Power BI, including dashboard creation, data visualization, and performance optimization. Hands-on experience was gained in Power Query for data cleaning and transformation, as well as in developing DAX measures and calculated columns. It also improved understanding of data modeling and building interactive dashboards.

-Analytical and Problem-Solving Skills

The project strengthened analytical thinking by identifying meaningful KPIs and interpreting business trends from raw data. It improved the ability to analyze patterns, validate insights, and solve data-related challenges effectively.

-Soft Skills and Team Collaboration

Working on this project improved communication, presentation, and time management skills. It also encouraged structured planning and collaborative discussion for refining dashboards and insights.

-Domain Knowledge and Application

The project provided practical exposure to food industry analytics, including customer behavior analysis, revenue trends, cuisine preferences, and the importance of quality consistency in restaurant performance.

6. Testimonials from team:-

Rekha Shri Giduthur

As the team leader for this project, I was responsible for planning the overall project workflow, coordinating tasks among team members, and ensuring timely completion of each milestone. I guided the team in understanding the project objectives, organizing the dataset, and structuring the analytical approach. I also contributed to data preparation, dashboard planning, and insight validation, ensuring that the visualizations clearly represented the analysis and supported meaningful conclusions. Throughout the project, I focused on maintaining effective communication within the team, resolving challenges, and ensuring that each member contributed effectively. Leading this project helped strengthen my leadership, collaboration, and problem-solving skills, while also providing valuable experience in managing a data analytics project from planning to final implementation. xsa

Rohit Ranjan

I was responsible for developing the Product Insights and Customer Segmentation dashboards in Power BI to analyze customer behavior and product performance. I created multiple visualizations including persona distribution charts, age group analysis, sentiment vs rating analysis, and purchase channel preference charts to understand how different customer segments interact with products and brands. By applying Power BI visualizations, DAX measures, and interactive slicers, I helped transform the dataset into meaningful insights that highlight customer loyalty patterns, demographic engagement, and purchasing behavior. The analysis supports businesses in identifying valuable customer segments and developing better customer retention strategies.

Nasina Asritha

As a team member of the Data Visualization project, I contributed to several important parts of the project. I actively participated in the data collection process, ensuring that the required dataset was gathered and organized properly. I also worked on designing the overall project flow to make the analysis and presentation structured and clear. In addition, I developed the project presentation using HTML, CSS, and JavaScript to create an interactive and visually appealing presentation. I also prepared the PowerPoint presentation used for the final demonstration. This project helped me improve my teamwork, data handling, and presentation development skills.

Naresh Kumar

The Product Insights Dashboard provides a clear and interactive view of product performance and customer preferences. The visualizations make it easy to understand key metrics such as brand performance, product quality ratings, and popular food items. This dashboard helps in making data-driven decisions and identifying opportunities to improve product strategy.

Nithya Thevath

The Strategy and Innovation Dashboard was developed as part of our project to better understand customer feedback. It uses sentiment analysis to classify comments as positive, neutral, or negative. The dashboard shows the total number of feedback responses and highlights the most common keywords used by customers. It also analyzes customer opinions about different product features such as taste, packaging, quality, and price. Additionally, it compares feedback across products and examines the relationship between customer ratings and sentiment. Overall, the dashboard helps us understand customer preferences and identify areas for product improvement.

Gummalla Nagalakshmi

Working on this project helped me gain practical experience in data analysis and customer behavior understanding. I learned how to analyze structured and unstructured data using tools like Power BI and apply techniques such as sentiment analysis and customer segmentation. This experience improved my analytical and problem-solving skills. Overall, the project strengthened my confidence in applying data analytics to real-world business problems.

Naitik Kanojiya

Using **VADER** (Valence Aware Dictionary and Sentiment Reasoner) via Python within Power Query, I classified customer feedback by calculating compound sentiment scores for products, features, and regions. Positive, neutral, or negative sentiment was determined by checking if the compound score was 0.05, between -0.05 and 0.05, or -0.05, respectively. I created a DAX measure for the positive feedback ratio using DIVIDE (CALCULATE ([Total Reviews], Sentiment="Positive"), [Total Reviews]) to visualize trends. Sentiment change over time was measured using a line chart, tracking the AVERAGE (Sentiment Score) over a date hierarchy.

10. Conclusion:-

1. Overall sentiment remains **positive (55%)**, with an **average score of (12%)**, indicating stable brand perception.

2.**Loyal customers** represent the largest segment and contribute significantly higher ratings, reinforcing str ong customer retention.

3. A small but important segment of at-**risk customers** requires targeted engagement strategies.

4. Product level sentiment differences highlights clear opportunities for Quality & Experience improvement.

5. Online **channels** and younger demographics drive higher engagement, suggesting **strong** digital **growth** potential.

1. Acknowledgements:-

We would like to express our heartfelt gratitude to:

Infosys Springboard Team – for organizing this valuable internship opportunity and providing a structured learning platform.

Our Mentor – for her continuous guidance, constructive feedback, and encouragement throughout the project.

My Teammates – for their collaboration, support, idea sharing, and teamwork during all phases of the project.

We also sincerely appreciate the opportunity to work on a real-world dataset that allowed us to apply analytical concepts in a practical environment. The exposure to Power BI, data modeling, and structured project execution has significantly enhanced our technical knowledge and problem-solving skills.

The collaborative environment, regular reviews, and milestone-based execution helped us develop not only technical expertise but also essential skills such as communication, accountability, time management, and professional coordination.

This internship has been a significant step in our professional growth and technical journey.